



中國海洋大學海德學院

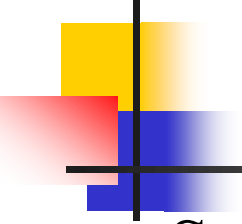
HAIDE COLLEGE, OCEAN UNIVERSITY OF CHINA

Statistical Modelling and Inference

Week 9

Outline:

- i. Estimation of parameters
- ii. Properties of residual's



Theorem's:

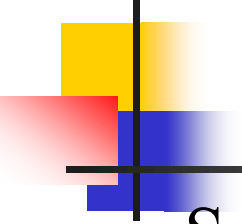
Suppose that $Y_1, Y_2, \dots, Y_n$ are independent with $E(Y_i) = \beta_0 + \beta_1 X_i$ and $Var(Y_i) = \sigma^2$, then show that:

i. $E(\hat{\beta}_0) = \beta_0$ and $E(\hat{\beta}_1) = \beta_1$

ii. $var(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}} \right)$

iii. $var(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$

iv. $cov(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{X}}{S_{xx}}$



Theorem's:

Suppose that $Y_1, Y_2, \dots, Y_n$ are independent with $E(Y_i) = \beta_0 + \beta_1 X_i$ and $Var(Y_i) = \sigma^2$

Then show that:

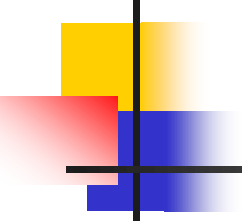
i. $E(\hat{\beta}_0 + \hat{\beta}_1 X_0) = \beta_0 + \beta_1 X_0$

ii. $Var(\hat{\beta}_0 + \hat{\beta}_1 X_0) = \sigma^2 \left(\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{S_{xx}} \right)$

Where:

$$E(\hat{\beta}_0) = \beta_0, E(\hat{\beta}_1) = \beta_1, Var(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}} \right)$$

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}} \quad \text{and} \quad Cov(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{X}}{S_{xx}}$$



Properties of Residuals (Error's)

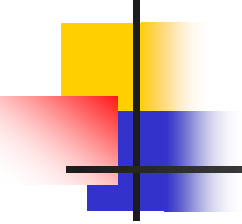
Properties of residual $\hat{e}_i = Y_i - \hat{Y}_i$

i. $\sum_{i=1}^n \hat{e}_i = 0$

ii. $\sum_{i=1}^n \hat{e}_i X_i = 0$

iii. $E(\hat{e}_i) = 0$

iv. $Var(\hat{e}_i) = \sigma^2 \left(1 - \frac{1}{n} - \frac{(X_i - \bar{X})^2}{S_{xx}} \right)$



Estimation of σ^2

To estimate σ^2 , we use the residual variance

$$s_e^2 = \frac{1}{n - 2} [S_{yy} - \hat{\beta}_1 S_{xy}]$$

Where $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ is the fitted and predicted value for the observation.

**Thank
You**

谢谢

